

Synthetic Label Generation — Disclosure

AI Chargeback Evidence Responder — Razorpay Buildathon Submission

This document discloses the exact hidden formula used to generate **outcome_true** (the XGBoost training label) in `golden_answers.csv`. This formula is used ONLY to create synthetic training data — it is never exposed to the model as a feature, and never used at inference/pipeline runtime. It exists solely so judges and evaluators can verify there is no label leakage and inspect exactly how ground truth was constructed.

1. Base Formula

```
base = 0.05

# Evidence signals (additive)
base += 0.22 if delivery_confirmed else 0.00
base += 0.18 if otp_auth_confirmed else 0.00
base += 0.07 if shipping_billing_match else 0.00
base += 0.07 if not prior_complaint_on_file else 0.00
base += 0.06 if days_to_dispute < 7 else 0.00

# Repeat-disputer penalty
base -= 0.05 * min(prior_dispute_count, 4)

# Amount effect
base += 0.06 if amount >= 10000
base += 0.03 if 5000 <= amount < 10000
base -= 0.02 if amount < 1000

# Item-category effect (centered around category mean)
category_weight = CATEGORY_ABUSE_WEIGHT[category]      # see Section 2
category_effect = (category_weight - CATEGORY_MEAN_WEIGHT) * 0.06
base += category_effect

# Reason-code-specific evidence effects
if reason_code == "unauthorized_transaction":
    base += 0.12 if otp_auth_confirmed else -0.10

elif reason_code == "item_not_received":
    base += 0.12 if delivery_confirmed else -0.15

elif reason_code == "not_as_described":
    base += 0.04 if not prior_complaint_on_file else -0.05

elif reason_code == "duplicate_charge":
    base += 0.08 if shipping_billing_match else 0.00
    base -= 0.04 * min(prior_dispute_count, 4)

# Noise (prevents a deterministic/unrealistic task)
noise ~ Normal(0, 0.06)
true_win_probability = clip(base + noise, 0.02, 0.98)

# Hidden ground-truth outcome
outcome_true ~ Bernoulli(true_win_probability)
```

2. Category Abuse-Weight Table

item_category	weight
apparel	1.3
electronics	1.2
digital_goods	1.1
home_goods	0.8
groceries	0.5

CATEGORY_MEAN_WEIGHT = mean of the above = 0.98

3. What XGBoost Actually Sees

Training features (X): amount, days_to_dispute, prior_dispute_count_at_time, delivery_confirmed, otp_auth_confirmed, shipping_billing_match, prior_complaint_on_file, one-hot reason_code, one-hot item_category.

Training label (y): outcome_true only.

Never used as a feature or label at any point: true_win_probability (the formula in Section 1). It is stored in golden_answers.csv strictly for evaluation and leakage-auditing purposes.

4. Reproducibility & Verification

- Random seed: SEED = 42 (numpy, random, Faker)
- Full generation logic: generate_data.py
- Held-out test metrics: ROC-AUC 0.6849, PR-AUC 0.6457 — consistent with a model learning genuine structure from noisy labels rather than leaking the generating formula (a leak would produce ROC-AUC in the high 0.90s)
- SHAP feature importance on the held-out test set independently recovers the two heaviest-weighted signals in the generator (delivery_confirmed, otp_auth_confirmed) as the top two predictors — discovered purely from noisy binary outcomes, not from access to the formula itself.

This synthetic dataset was constructed because no real merchant dispute-outcome data was available to buildathon participants. The generation logic above is provided in full for transparency and independent verification.